

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

ACHERA

MARKING SCHEME

NOVEMBER 2014

CHEMISTRY

9189/4

1 (a) (i) the (number and) sequence of amino acids in a polypeptide chain; *AW* [1]

(ii) hydrogen bonds in a secondary structure are formed between the N-H and C=O groups about the amide links while; *(1)*
~~amino acid~~ *carbonyl of amide*

Hydrogen bonds in a tertiary structure are formed between the polar Groups found in the R-groups of the amino acids; *structure naming* [1]
 OH / NH_2
 $\text{R} \text{COOH}$

(b) (i) hydrophobic means water heating/ *water repellent* low solubility in water; [1]

(ii) 1. The non-polar groups in glycine and alanine confers hydrophobicity; *contain water hating R groups* [1] *(CH₃)*
 2. In glycine the R-group is a hydrogen which has least contribution to hydrophobic interactions. *AW* *(1)* *(2)*

(c) (i) They have similar properties because both are polyamides/both have the amide link; *Accept formula (reject peptide linkage)* [1]

(ii) possible uses: making ropes/nets/fishing line/hats/fabric (socks; gloves); *sent* [1]

(d) It depends on the number of glycine/alanine combined poly-alanine/ poly glycine-alanine (type of polymer); different species of spiders; [1]

[Total: 10]

2

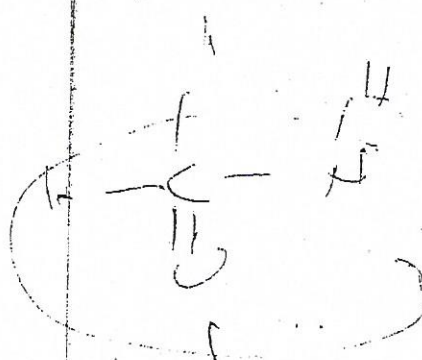
(a) process transcription [1]
 site occurs in the nucleus [1]

process translation [1]
 site occurs in the cytoplasm */ribosome* [1]

(b) (i) Gene – is the section of DNA that is transcribed/copied to give the information needed to make one polypeptide chain. *AW* [1]

(ii) Two genes are needed; one for the α -helical chain and one for the one β -pleated chain. [1]

(iii) The base sequence on a gene is the one that is copied out producing a */complementary* mRNA molecule. [1]



- (iv) Cofactor in haemoglobin is Fe^{2+}

Function: Bind oxygen thereby facilitating oxygen transport to body tissues *4m / transport oxy*

[1]

- (v) carbon monoxide bind haemoglobin irreversibly thereby reducing oxygen transportation *penetration*

[1]

[Total: 10]

3

- (a) Transport proteins; *channel proteins / carrier protein*
control passage of substances into and out of cells;

[1]

[1]

glycoproteins/receptor proteins; *transmission of nerve impulses*
provide signals that alters reactions occurring inside the cell or inside the membrane; *receptor site*

[1]

[1]

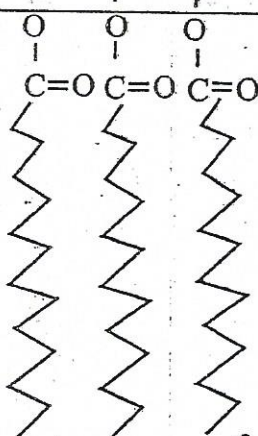
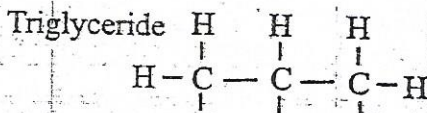
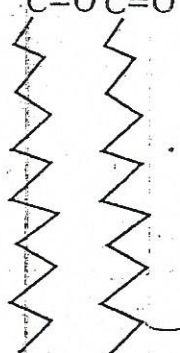
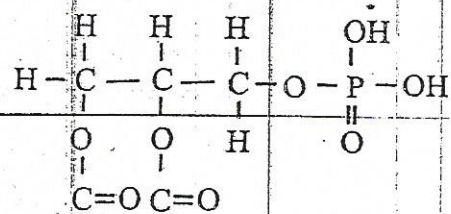
Or

Catalytic proteins – participate as enzymes in reactions occurring in the membrane; *1*

1

(b)

Phosphoglyceride

*1*Accept *be* defined - R*1*

long hydrocarbon chain
fat and

Phosphoglyceride has a phosphate group;

check down

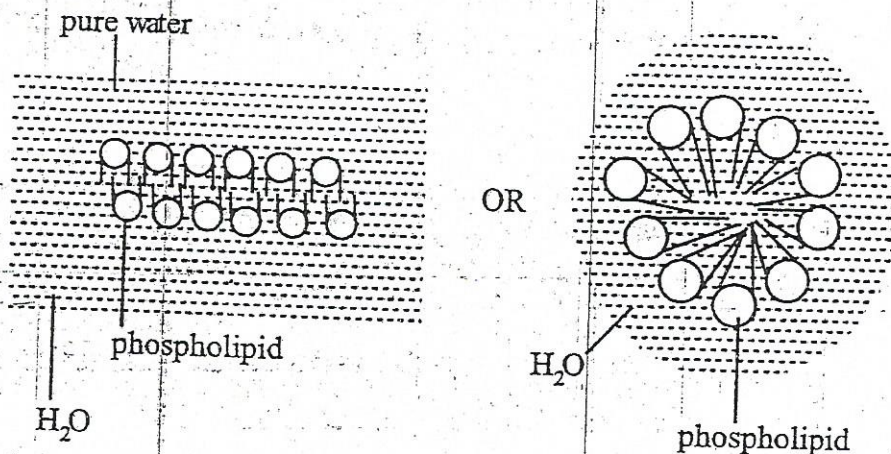
[1]

Drawing without def.

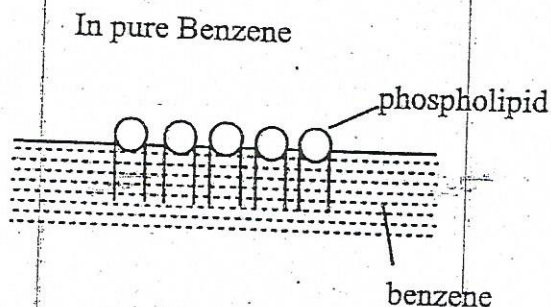
R -

1 mark instead of 2
low both

(c) (i)

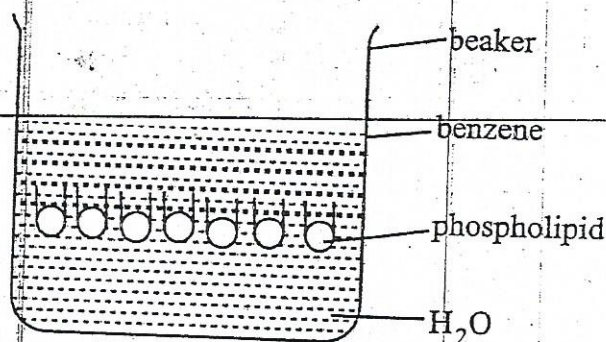


(ii)



Accept without labels

(iii)



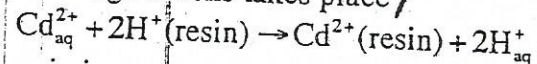
4

(a) (i)

Waste water passed through the ion-exchange resin;

exchange resin;

exchange of ions takes place /



resin is regenerated;

/ H^+

(ii)

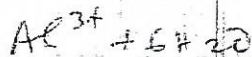
Insoluble compounds/immobilised groups
Polymer with polar groups/
Bind ions reversibly;

[1]

[1]

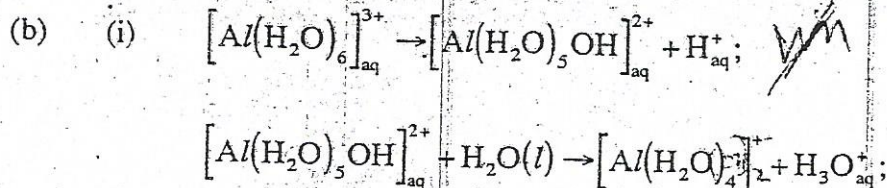
[1]

[1]



5

Accept



(1)

(3)

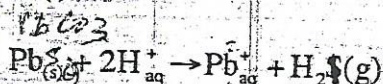
[1]

H^+ ions released increases soil acidity;

[1]

(ii) Pb_3 dissolves in acidic soil solution to give free Pb^{2+} ions;

[1]



Pb_3 exist as a solid Pbs ;

silicates (2)

exchange (1) eqn (1)

(1)

5

(a) (i) Suspended matter settles;
organic impurities are oxidised;

killing bacteria/microorganism

[1]

[1]

(ii) Forms a precipitate of aluminium hydroxide; / flocculent / impurities - (1)
Hydroxide adsorbs suspended matter and sinks to the bottom;

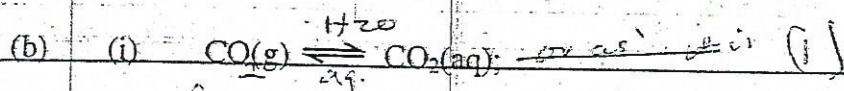
[1]

(iii) To destroy bacteria; / AW

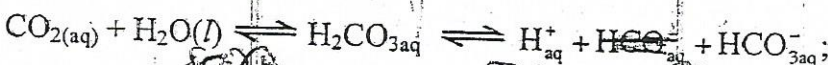
[1]



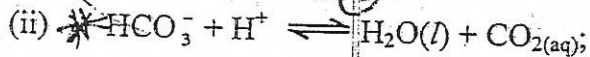
[1]



[1]



[1]



[1]

When algae absorbs CO_2 , the equilibrium shifts to the right thus reducing acidity.

[1]

or 2

(a) (i) $[X] = \left(\frac{V_{aq}}{V_{org}K + V_{aq}} \right) [X]_0;$ [1]

$[X]$ = final concentration of X
 $[X]_0 = 0.05 \text{ moldm}^{-3}$ [1]

$V_{aq} = V_{org} = 25 \text{ ml}$

$K = 9.6$

Thus $[X] = \left(\frac{25 \text{ ml}}{25 \times 9.6 + 25} \right) (0.05) \text{ moldm}^{-3}$
 $= 0.00472 \text{ moldm}^{-3}$ [1]

(ii) $[X] = \left(\frac{V_{aq}}{V_{org}K + V_{aq}} \right)^2 [X]_0;$ [1]

$[X]$ = final concentration

2 = number of extractions

$V_{org} = 12.5 \text{ ml}, V_{aq} = 25 \text{ ml}$ [1]

$[X] = \left(\frac{25}{12.5 \times 9.6 + 25} \right)^2 (0.05) \text{ moldm}^{-3}$
 $= 0.00149 \text{ moldm}^{-3}$ [1]

(b) Larger amount extracted with two 12.5 ml portions [1]

(c) (i) Distribution constant (K) increases, *(extract ↑)* [1]
~~↑~~ because dissolving a salt (NaCl) in water makes H₂O more polar. [1]

(ii) Distribution constant varies with temperature. *and concentration of* [1]
 [Total: 10]

(a) (i) 1. alanine will have the largest R_f value; [1]

because its least polar / ~~least~~ *highest* molecular weight. [1]

2. aspartic acid will have the smallest R_f value; [1]

because its most polar/doubly charged [1]

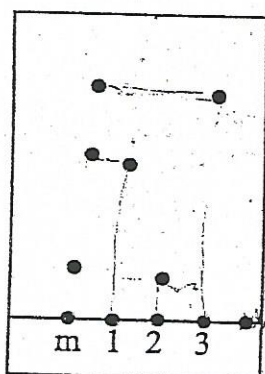
L = no x
solved
 $\frac{40}{20} = 2$
 $\frac{2}{20}$

- (ii) Ag^+ ion form dative bonds/complexes the $-\text{NH}_2$ groups
thus R_f value decreases

[1]

[1]

(b)



m = mixture
1 = serine,
2 = aspartic acid
3 = alanine,

[2 marks for all labels]

- (c) separation of dyes/drugs/DNA/products of nitration/blood serum component.

[1]

- (a) (i) Boiling point of H_2O is lowered;

[1]

Reason: H_2O and CCl_4 are immiscible;

[1]

each component produces its own vapour pressure independent of the other; / vapour pressure $\uparrow$

so $P_{\text{H}_2\text{O}} + P_{\text{CCl}_4} = 1$ atmosphere at a temperature far much lower than boiling point of H_2O or CCl_4 ;

- (ii) Boiling point of mixture decreases;

[1]

because boiling takes place when vapour pressure equals to atmospheric pressure;

- (b) (i) method: solvent extraction

[1]

- (ii) In a low pressure environment;

[1]

steam distillation can be performed at very low temperature;

[1]

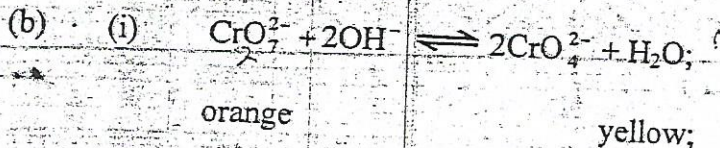
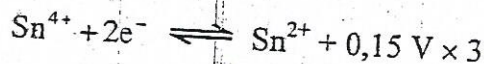
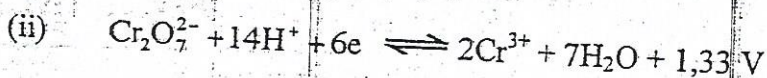
preventing thermal decomposition of the perfume;

[1]

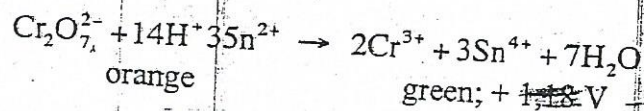
[Total: 10]

10

- (a) (i) - improves strength of steel; *hardening*
 - makes steel resistant to corrosion;
 - decoration; *chromium*
- (ii) Cr^{3+} bonds with water molecules through dative bonding;
 Bond angle 90° ;

[1]
[1]
[1][1]
[1]

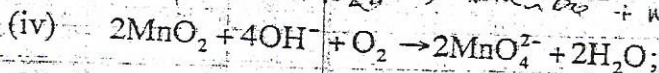
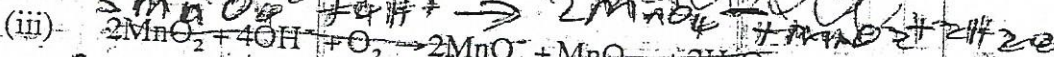
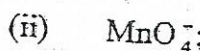
[1]

[1]
[1]

[Total: 10]

11

- (a) (i) Green;



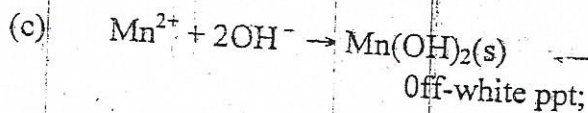
[1]

[1]

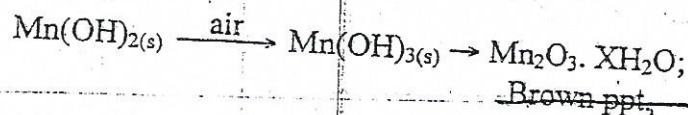
[1]

- (b) - catalyst;
 - depolariser in dry cells;

Lab preparation of chlorine

[1]
[1][1]
[1]

On standing a brown ppt is observed

[1]
[1]

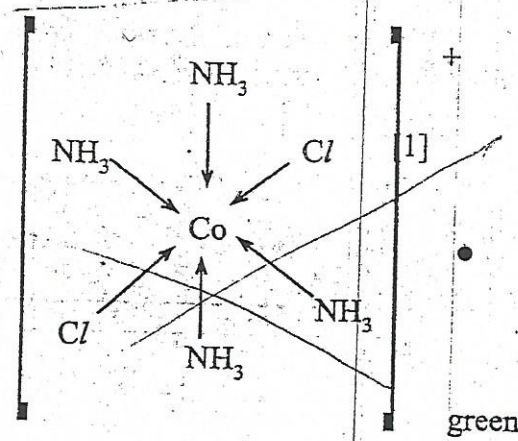
[Total: 10]

12

(a)

(i)

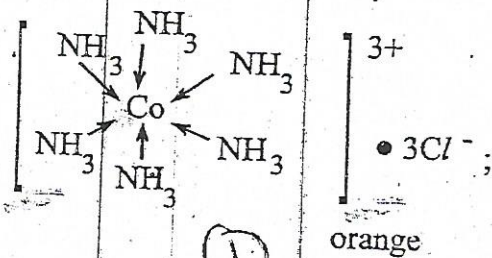
Ionisation:



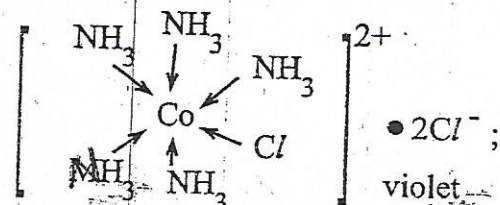
green

Some present.

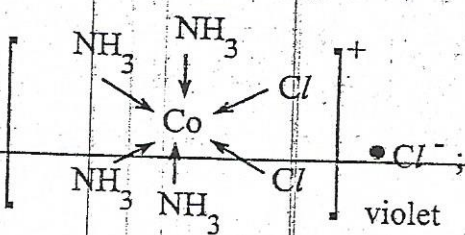
(ii)



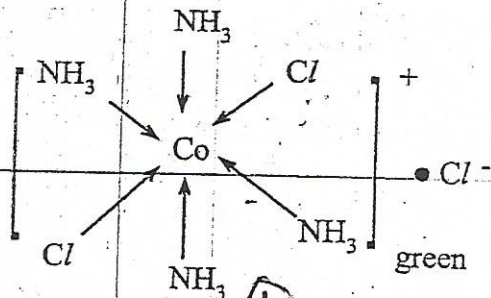
orange



violet



violet



green

(iii) $\text{CoCl}(\text{NH}_3)_6$ - no dipole moment symmetrical distribution of ligands $\rightarrow$

$\text{CoCl}_3(\text{NH}_3)_5$ - dipole moment present

Non symmetrical distribution of ligands

$\text{CoCl}_3(\text{NH}_3)_4$ (violet)

Dipole/moment present
non-symmetrical distribution
of ligands

[1]

[1]

$\text{CoCl}_3(\text{NH}_3)_4$ (green).

- no dipole moment;
- symmetrical distribution of ligands;

- (b) (i) anaemia;
- (ii) red blood cells;

WMA
[1]

[1]

[1]